Mid-Term Feedback

A. **Instructions (1 Unclear -10 Very Clear): 9**

B. **Questions (1 Unclear -10 Very Clear): 9**

C. **Difficulty (1 Very Difficult – 10 Very Easy): 5.5 Moderate**

D. **Time (1 Very less - 10 Plenty): 6**

MCQ for endsem, no negative marking, home preferred, cheat sheet not allowed, discuss answers, partial marking, self brought calculator

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Singling Out on the Singular Matrix

Sir if p >> n, the matrix is not square.

So how can we comment on it being singular?

sir one of the reason it could if rows/columns is become

linearly dependent when P is large kind of over-parameterize .

then it can become singular

Singling Out on the Singular Matrix

A singular matrix is one for which inverse does not exist

- Further singularity applies only to square matrices
- By definition, a matrix is singular if det(A)=0.
- Calculating the inverse of a matrix requires dividing by its determinant (A^{-1}=adj(A)/det(A)), division by zero is undefined, making it impossible for an inverse to exist.

Singularity applies only to square matrices

- Concepts of determinants & standard matrix inverses are strictly defined only for square matrices ($n \times n$).
- Since non-square matrices do not have a determinant, they cannot be classified as singular or non-singular

How can a Rectangular Matrix Be(come) Singular?

While a rectangular design matrix X technically cannot be "singular" in the literal sense (since singularity is mathematically reserved for square matrices), it exhibits the **exact equivalent of singularity** in regression.

In linear regression, this equivalent condition is called being **rank-deficient**, which triggers actual singularity in the underlying math. Here is **how** and **when** it happens.

1. How It Happens (The Mathematical "How")

In ordinary least squares (OLS) linear regression, the goal is to solve for the vector of coefficients ($\hat{\beta}$) using the **normal equations**:

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

How can a Rectangular Matrix Be(come) Singular?

Here is the chain reaction:

1. X is a rectangular matrix of size $n \times p$ (where n is the number of observations/rows, and p is the number of features/columns).
2. To find the regression coefficients, the computer must compute $X^T X$, which is a **square matrix** of size $p \times p$.
3. If the rectangular matrix X is **rank-deficient** (meaning its columns are not completely independent), the square matrix $X^T X$ becomes **perfectly singular** ($\det(X^T X) = 0$).
4. Because $X^T X$ is singular, **it cannot be inverted**. The software crashes or outputs errors

When can a Rectangular Matrix Be(come) Singular?

2. When It Happens (The Statistical "When")

A rectangular matrix will lose its full rank and cause singularity in two primary scenarios:

Scenario A: Perfect Multicollinearity (Redundant Features)

This occurs when one or more columns in your dataset can be perfectly predicted by a linear combination of the other columns. DataCamp

- **The Dummy Variable Trap:** If you have a categorical variable like "Gender" (Male/Female) and you include an intercept column, a "Male" column, *and* a "Female" column, the matrix becomes singular. This is because $\text{Male} + \text{Female} = 1$ (the intercept). One column is mathematically redundant.
- **Duplicate or Scaled Features:** Including the same data in different units—such as a column for `Weight_in_Pounds` and another for `Weight_in_Kilograms`. Since one is just a scalar multiple of the other ($1 \text{ kg} = 2.2 \text{ lbs}$), the columns are linearly dependent.
- **Sum Constraints:** If you include features that always add up to a constant total (e.g., `Research_Budget`, `Marketing_Budget`, and `Operations_Budget` where the three always equal `Total_Budget`), the columns will be linearly dependent.

When can a Rectangular Matrix Be(come) Singular?

Scenario B: The "High-Dimensional" Problem ($p > n$)

This happens when you try to estimate more parameters than you have observations.

- If you have 100 features ($p = 100$) but only 50 rows of data ($n = 50$), the maximum possible rank of your rectangular matrix X is capped at 50.
- Because the rank (50) is less than the number of features (100), the columns are inherently forced to be linearly dependent. The square matrix $X^T X$ will automatically be singular.

How It Is Fixed

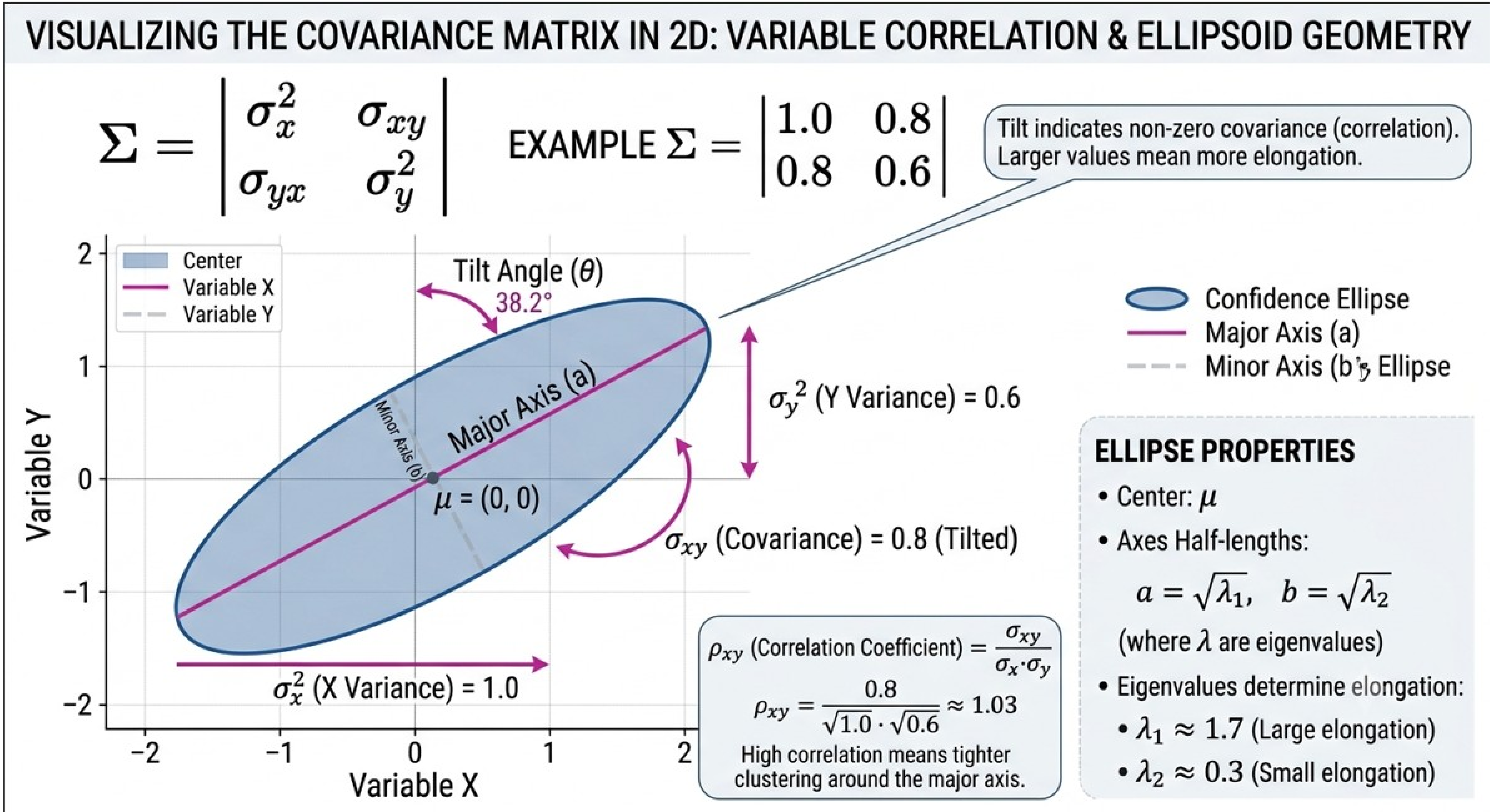
When a rectangular matrix faces this issue, data scientists bypass it by dropping the redundant features, gathering more data rows, or using **Regularization (Ridge/Lasso)**, which artificially adds a small value to the diagonal of $X^T X$ to force it to become non-singular and invertible.

DataCamp

Too Much Correlation

"On the 'Demonstration of Variance-Covariance Principles' slide (Slide No. 2 of VarCovProblems_pyLinks.pptx), the sample problem lists a correlation coefficient value of '1.03'. If the correlation coefficient is equal to or greater than 1 ((>=1)), it should form a straight line rather than an ellipse, right? Could you kindly check this sample problem from the shared slide? Thanks in advance”

Demonstration of Variance-Covariance Principles



1. Mathematical proof (using the Cauchy–Schwarz inequality)

The sample correlation coefficient is

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

Define the centered vectors

$$\mathbf{x} = \begin{bmatrix} x_1 - \bar{x} \\ x_2 - \bar{x} \\ \vdots \\ x_n - \bar{x} \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} y_1 - \bar{y} \\ y_2 - \bar{y} \\ \vdots \\ y_n - \bar{y} \end{bmatrix}.$$

Then

$$r = \frac{\mathbf{x}^T\mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|}.$$

The **Cauchy–Schwarz inequality** states that

$$|\mathbf{x}^T\mathbf{y}| \le \|\mathbf{x}\|\,\|\mathbf{y}\|.$$

Dividing both sides by the positive denominator,

$$|r| = \frac{|\mathbf{x}^T\mathbf{y}|}{\|\mathbf{x}\|\|\mathbf{y}\|} \le 1.$$

Hence,

$$-1 \le r \le 1.$$

So a correlation greater than 1 would violate a fundamental inequality of vector algebra.

Covariance and Collinearity

Doubt :

Sir, when we say we need to maximize the covariance of **X** (with **Y**) in-order to find latent variable(s)

so here are we ignoring the collinearity in PLS as collinearity will be high in this case ?

Covariance and Collinearity

Rephrasing the question:

By maximizing covariance between **X** and **Y** (scores), **Y** becomes strongly related to **X** and hence *'collinear'* with **Y**. Does this mean we are ignoring collinearity between **X** and **Y** as collinearity will be high in this case?

X and **Y** are unchanged. There may be strong collinearity within **X** which makes linear regression infeasible.

Even in linear regression, strong strength of linear relationship (correlation) between **X** and **Y** is welcomed so that a good fit is achieved.

You put all **X** in one matrix but **X** and **Y** are separate matrices

Just as X'X matrix become singular due to strong collinearity why can't X'Y matrix become singular due to high correlation?

This is an excellent question. The key point is that **singularity is defined only for square matrices**.

1. Why can $X^T X$ become singular?

Suppose X is an $n \times p$ predictor matrix.

Then

$$X^T X$$

is a $p \times p$ **square matrix.**

If two or more columns of X are linearly dependent (perfect multicollinearity), then

$$\operatorname{rank}(X) < p,$$

and consequently,

$$\det(X^T X) = 0.$$

Thus $X^T X$ is singular and cannot be inverted.

This is why ordinary least squares fails.

Covariance and Collinearity

2. What about X^TY?

If Y is a single response,

$$Y \text{ is } n \times 1,$$

so

$$X^TY$$

is a $p \times 1$ **vector**, not a square matrix.

A vector **cannot be singular or nonsingular** because singularity is defined only for square matrices

Even if X and Y are perfectly correlated,

$$X^TY$$

simply contains large inner products. Nothing becomes non-invertible.

3. If there are multiple responses?

Suppose

$$Y : n \times k.$$

Then

$$X^T Y$$

is a $p \times k$ matrix.

Again, unless $p = k$, it is not even square.

Even if it happens to be square ($p = k$), the invertibility of $X^T Y$ is **not** what determines whether regression can be solved. Ordinary least squares never requires inverting $X^T Y$.

Instead, the regression coefficients are

$$\hat{B} = (X^T X)^{-1} X^T Y.$$

The only matrix that must be inverted is $X^T X$.

https://chatgpt.com/c/6a5e3f65-a90c-83ee-88d0-f7041512dfa4

A useful rule

- **Among predictors (X-X)** → use **collinearity** or **multicollinearity.**
- **Between predictors and response (X-Y)** → use **correlation**, **association**, or **linear relationship.**

Recommended wording

Instead of:

> "PLS maximizes the collinearity between X and Y."

say:

> **"PLS maximizes the covariance (or correlation) between the latent scores derived from X and Y."**

or

> **"PLS seeks latent variables from X that have the strongest linear relationship with the latent variables from Y."**

These are the standard and technically correct ways to describe what PLS is doing.

Covariance and Collinearity

X and **Y** are unchanged. There may be strong collinearity within **X** which makes linear regression infeasible. In PLS we are talking about (cor)relationship *between* **X** and **Y**.

As told in class we are finding relation between **X** and **Y** through **T** i.e., we express **Y** in terms of **T**.

While **X** may be collinear, the components of **T** ($\mathbf{t}_1, \mathbf{t}_2, \ldots \mathbf{t}_a$) are orthogonal. Hence the **T** matrix does not suffer from singularity issues when you related **T** to **Y**.

PLS transforms

$$X \to T$$

where

$$T = [t_1, t_2, \dots]$$

The score vectors satisfy

$$t_i^T t_j = 0, \qquad i \neq j.$$

Thus the latent variables are orthogonal.

Instead of regressing on highly correlated original variables

$$X_1, X_2, \dots, X_p,$$

we regress on uncorrelated scores

$$t_1, t_2, \dots$$

which eliminates the numerical instability caused by multicollinearity.

PLS also forms

$$u_1, u_2, \ldots$$

from Y.

The first pair (t_1, u_1) has maximum covariance.

After extracting this pair, both X and Y are deflated, and the next pair is computed from the residual information.

So PLS does **not** simply make everything highly correlated; it successively extracts independent latent dimensions that explain the shared variation.

Relationship
Collinearity among X variables
Correlation between X and Y
Correlation among latent X scores
Correlation between each latent X score and its corresponding latent Y score

So, although the extracted score t is made highly correlated with the corresponding response score u, the original collinearity among the predictors has been transformed into a stable set of orthogonal latent variables. This is why PLS can successfully handle highly collinear predictor data while still maximizing predictive power.

Relationship	Desired?	Why?
Collinearity among X variables	**No**	Causes unstable regression coefficients
Correlation between X and Y	**Yes**	Enables accurate prediction
Correlation among latent X scores	**No**	Scores are orthogonal after extraction
Correlation between each latent X score and its corresponding latent Y score	**Yes**	This is the optimization objective of PLS

So, although the extracted score t is made highly correlated with the corresponding response score u, the original collinearity among the predictors has been transformed into a stable set of orthogonal latent variables. This is why PLS can successfully handle highly collinear predictor data while still maximizing predictive power.

Covariance and Collinearity

Rephrasing the question:

By maximizing covariance between **X** and **Y** (scores), **Y** becomes strongly related to **X** and hence collinear with **Y**. Does this mean we are ignoring collinearity between **X** and **Y** as collinearity will be high in this case?

Answer:

In PLS we want to find suitable directions for the factors/latent variables/components.

We maximize covariance between the **X** and **Y** scores. **T=Xw** and **U=Yq**

We find (principal) components in PCA that maximize the variance in **X** alone.

In PLS we find the components by maximizing the covariance involving **X** and **Y**.

In PCA you have no **Y** just **X.**

In PLS and PCA you find directions of maximum variance or covariance. IN PLS the **direction of the latent variables (factors) identified** is based on maximizing covariance between **X** and **Y** which gives PLS its predictive power.

a

Covariance and Collinearity

Rephrasing the question:

Then how and where does PLS2 maximize the covariance between X and Y ?

Answer:

Taking a step back we find (principal) components in PCA that maximize the variance in **X** alone.

In PLS we find the components by maximizing the covariance involving **X** and **Y**.

In PCA you have no **Y** just **X.**

In PLS and PCA you find directions of maximum variance or covariance. IN PLS the **direction of the latent variables (factors) identified** is based on maximizing covariance between **X** and **Y** which gives PLS its predictive power.